

Prolongation, projection, and the quantifier discipline: an algorithm for the ansatz-plus-PDE solution space and the resolution of the $v = v_1x + v_2y + v_3z$ puzzle

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Abstract

A companion note to *A New Method of Solving PDEs*. Given an ansatz $A(c)$ (a differentially triangular family with constant parameters c) and a PDE P , we survey the algorithm families available for computing the solution space of the combined problem, then develop a proposed three-phase algorithm: prolong $A \cup \{P\}$ to differential closure while tracking the inequations actually used (Phase I), recurse Thomas-style on the tracked inequation loci (Phase II), and project each resulting system onto the constants under the quantifier discipline *independent variables universal, dependent variables and constants existential* (Phase III). The intellectual center of the note is a puzzle: applied naively, the universal coefficient-zeroing of Phase III would read the ansatz-defining equation $v = v_1x + v_2y + v_3z$ as forcing $v_1 = v_2 = v_3 = 0$. We resolve the puzzle by making the quantifier discipline precise. Universal quantification is implemented by *coefficient collection* and is sound only over symbols that are algebraically free of one another — which the fully reduced remainder’s variables are, and a defining equation’s leader is not; existential quantification is implemented by *Gröbner elimination* and is sound only *after* differential closure (prolongation to passivity). The naive Phase III interleaves the two operations in an order in which they do not commute, and we exhibit small examples where it over-constrains (the v -equation) and where it under-constrains (returning $\mathbb{C}$ where the correct existence locus is $\{c = 1\}$). Corrected, the algorithm computes the per-cell *existence* locus — the joint-variety projection $\mathcal{V} \cup \{\text{partial strata}\}$ — with the *membership* locus $\mathcal{V}$ selectable per cell by flipping the quantifier on the parametric derivatives, equivalently by the dimension-preservation criterion.

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1 Problem statement: two loci, not one

Let $A(c) = \{a_1, \dots, a_k\}$ be the *ansatz*: a differentially triangular set of differential polynomials in dependent variables $\Psi, v, \dots$ over independent variables x, y, z (derivations $\partial_x, \partial_y, \partial_z$), whose coefficients involve constant parameters $c \in \mathbb{C}^n$. Let P be the PDE. The running concrete case is the hydrogen ansatz of the paper [1]:

$$\begin{aligned}
& (a_0 + a_1v) \Psi'' + (b_0 + b_1v) \Psi' + (c_0 + c_1v) \Psi = 0, \\
& v - (v_1x + v_2y + v_3z + v_4r) = 0, \quad r^2 - (x^2 + y^2 + z^2) = 0, \\
& \Psi_x - \Psi'v_x = 0, \quad \Psi_y - \Psi'v_y = 0, \quad \Psi_z - \Psi'v_z = 0,
\end{aligned} \tag{1}$$

with P the Schrödinger equation $-\frac{1}{2}(\Psi_{xx} + \Psi_{yy} + \Psi_{zz}) - \frac{1}{r}\Psi = E\Psi$ and eleven constants $c = (E, v_1, \dots, v_4, a_0, a_1, b_0, b_1, c_0, c_1)$.

“The solution space of ansatz + PDE” admits two readings, and the entire subject of this note turns on keeping them distinct [2]:

$$\mathcal{M} = \{c : P \in [A(c)]\} \quad (\text{membership locus: every ansatz member solves } P), \tag{2}$$

$$\mathcal{E} = \pi_c(\text{Sol}(A) \cap \text{Sol}(P)) \setminus \{\text{trivial}\} \quad (\text{existence locus: some nonzero member solves } P). \tag{3}$$

Here $[A(c)]$ is the differential ideal generated by the ansatz at c , and “trivial” excludes the identically-zero solution, which for the linear-homogeneous class solves everything at every c and must be quotiented away before either locus is informative [3, §2bis, “Two saturations”].

$\mathcal{M}$ is a *universal* condition over the ansatz family; $\mathcal{E}$ is *existential*. Always $\mathcal{M} \subseteq \mathcal{E}$, and the difference

$$\mathcal{E} \setminus \mathcal{M} = \{\text{partial-solution strata}\}$$

consists of parameter values where a *proper nontrivial subfamily* of the ansatz solves P while the family as a whole does not [2, Def. 1]. The paper’s Algorithm 1 (reduce P modulo A , project the remainder onto the constants) computes $\mathcal{M}$; decomposing the combined system $A \cup \{P\}$ and projecting computes $\mathcal{E}$. The two-line example

$$A : \Psi'' - c\Psi, \quad P : \Psi' - \Psi$$

separates them: $\mathcal{M} = \emptyset$ (no c makes the whole two-dimensional family $K_1 e^{\omega x} + K_2 e^{-\omega x}$ obey $\Psi' = \Psi$), yet $\mathcal{E} = \{c = 1\}$ (the subfamily $K e^x$ solves at $c = 1$) [2, Ex. 1].

The paper’s own motivation for treating A and P *on an equal footing* — one combined decomposition rather than reduce-then-project — is precisely that the partial strata are invisible to the membership formulation by construction, and that some of them carry physically meaningful solutions: in the hydrogen differential Thomas run, the bad-locus cell at $E = -1/8$ is the hydrogen $2s$ state $(1 - r/2)e^{-r/2}$, whose constants provably fail the projection system of Algorithm 1 [6]. Any algorithm that adjoins P to A must therefore be judged against *both* targets: which locus does it compute, and can it be tuned to compute the other?

2 Survey: the algorithm families

Four routes to the ansatz-plus-PDE solution space are on the table. The conceptual comparison — what question each answers and what triggers a case split — is developed at length in [4]; we summarize, adding only what this note needs. Prior-art verdicts and full citations are in the litsearch report [5].

2.1 Reduce-then-project (the paper’s Algorithm 1)

Ritt’s full reduction of P modulo A over the field $\mathbb{Q}(c)$ produces $h \cdot P - r \in [A]$ with r fully reduced and h a power product of initials and separants of A (the “denominator”). The projection step treats each surviving non-constant symbol as an independent indeterminate, collects

$$r = \sum_{\alpha=1}^N m_{\alpha} p_{\alpha}(c), \quad (4)$$

m_{α} distinct power products in the non-constants, and returns $\mathcal{V} = V(p_1, \dots, p_N)$ as minimal primes (commutative GTZ). For hydrogen the collection runs over exactly x, y, z, r, Ψ, Ψ' [1, §Projection Step], yields 28 constant equations, and the five minimal primes reproduced by `joca.sage`.

Question answered: the membership locus, $\mathcal{V} = \mathcal{M}$ (on the locus $h \neq 0$; Theorem 1’s completeness hypothesis). *Splits:* none in the good case — one reduction, one GTZ call; the general Algorithm 2 recurses only on the *bad locus* where the hypotheses of the completeness theorem fail. *Cost/faithfulness:* the constants never enter the differential engine (they live in the coefficient field), which is the practical lever — and the price is that the bad-locus strata (e.g. $a_0 = a_1 = 0$, carrying the $1s$ and $2s$ states as first-order-ODE solutions) are outside the guarantee.

The closest published relative is Chaharbashloo–Basiri–Rahmany–Zarrinkamar [20]: substitute an *explicit* physics ansatz $\psi = g(r)e^{y(r)}$ into the wave equation, equate powers of r by hand, and solve the resulting constant system by a commutative Gröbner basis. That is exactly a reduce-then-project computation — and, as §5 will make precise, the reason their raw power-equating is sound is that an explicit ansatz has no defining equations left: substitution has already eliminated every dependent leader.

2.2 Differential Thomas decomposition

Bächler–Gerdt–Lange–Hegermann–Robertz [8] (implementation for PDE systems: the MAPLE package TDDS [9]; project Sage port in progress, `joca-thomas-native-dt.sage` / `thomas_cells.py`). Decomposes any system into finitely many *disjoint* simple systems (cells): each differentially triangular, *passive* (Janet-involutive: every non-multiplicative prolongation reduces to zero), with attached inequations (initials, separants, squarefreeness witnesses).

Question answered: the complete, disjoint stratification of $\text{Sol}(A \cup \{P\})$ — the existential object; the constant-space shadow of the cells is $\mathcal{E}$ together with its case structure. With the constants modeled as $\dot{c}_i = 0$, the constant-only inequations become parameter-degeneracy conditions and the decomposition is automatically comprehensive over the constants. *Splits:* wherever the *description* degenerates — every initial, separant, or discriminant of every equation processed during completion, whether or not its vanishing changes anything about existence. *Status here:* the open-maple run of the hydrogen combined system completes with 29 cells; the bad-locus cells 26/27 carry the $E = -1/2$ and $E = -1/8$ strata [6].

2.3 Parametric Rosenfeld–Gröbner

Fakouri–Rahmany–Basiri [12]; the non-parametric engine is Boulrier’s BLAD/BMI implementing Rosenfeld–Gröbner [11]. Regular differential chains represent $[C] : H_C^\infty$; a comprehensive-Gröbner-style bookkeeping splits parameter space wherever an initial or separant’s vanishing is ambiguous mid-elimination.

Question answered: per parameter cell, a regular-chain description of the (saturated) solutions — existential in character, but of the *saturation* $[C] : H_C^\infty$, so strata inside initial/separant loci are systematically divided out rather than kept. *Splits:* parameter conditions arising during elimination. *Status here:* the project port (`~/parametric-rg`) reproduces the paper’s examples; on hydrogen the parameter branching is cheap but the differential reduction hits the BLAD coefficient-swell wall [4]. Saturation-unfaithfulness is not hypothetical: reducing the PDE against the RG chain’s bare equations re-admits the spurious prime (a_0, a_1) while absorbing (losing) the two genuine strata inside it [7, `joca-rg.sage` discussion].

2.4 Bounded prolongation to the involutive order, then pure commutative algebra

The route developed in [3]: prolong $A \cup \{P\}$ to an order N at which the system is involutive, treat every jet of order $\leq N$ as an independent algebraic indeterminate, form the *un-saturated* polynomial ideal $J = ((A \cup \{P\})^{[N]}) \subseteq K[\text{jets}_{\leq N}, x, y, z, r, c]$, and finish with any commutative-algebra algorithm. The differential engine is touched exactly once. The semantic core is published (Lange-Hegemann’s Theorem 4.1 [10]: truncated solution sets of an involutive system equal the solution sets of its algebraized truncation; the construction itself appears inside Golubitsky–Kondratieva–Ovchinnikov’s treatment of the generalized Ritt problem [14], there with the bad-locus-deleting saturation : H_C^∞ that this route deliberately omits). Comprehensive analogues on the algebraic side: comprehensive involutive systems [15] and comprehensive triangular decomposition [16, 17].

Question answered: the existence locus, $V((J : \Psi^\infty) \cap K[c]) = \bar{\mathcal{E}}$ — the $: \Psi^\infty$ quotient (delete only the zero section) is mandatory, and is a different operation from the H_A -saturation the route avoids [3, §2bis]. *Splits:* optional — a constructible (algebraic-Thomas) split on initials/separants/discriminants recovers the disjoint strata; the single-ideal route returns the Zariski closure, a superset that never loses a component. *Two corrections from the July 2026 implementation attempt* [6], both inherited by the algorithm of §3:

- The involutive order of the *ansatz alone* ($N_0(A) = 2$ for hydrogen, from the single- v structure) is **not** the involutive order of the *combined* system: the PDE–ODE critical pair (shared leader Ψ'') feeds conditions $D^k \rho$ (derivatives of the Ritt remainder) into the ideal only at order $2 + k$. No a-priori $N_0(A \cup \{P\})$ is known; *prolong-until-stable* — watch the elimination ideal $(J^{(N)} : \Psi^\infty) \cap K[c]$ ascend and stop at stabilization (Hilbert basis theorem) — is the algorithm, not a safety check.

- The direct implementation over all eleven constants as ring variables hits a comprehensive-elimination cliff (no completion even over GF(32003) at $N = 2$); per-cell staging over the already-computed Thomas cells, or pointwise Nullstellensatz probes, are the practical fallbacks.

2.5 Summary table

	question answered	what triggers a split	principal limitation
Algorithm 1 (reduce+project)	membership $\mathcal{M}$ on $\{h \neq 0\}$	only bad-locus recursion (Alg. 2)	misses partial strata and the $h = 0$ strata by construction
differential Thomas	full disjoint stratification of $\text{Sol}(A \cup \{P\})$; shadow = $\mathcal{E}$	every initial / separant / discriminant met during completion	splitting cost; constants ride through the differential engine
parametric RG	per-cell regular chains of the <i>saturated</i> ideal	parameter ambiguity mid-elimination	saturation deletes in-locus strata; coefficient swell
bounded prolongation	$\overline{\mathcal{E}}$ (closure), strata optional	optional constructible split	no a-priori $N_0(A \cup \{P\})$; parametric elimination cliff

3 The proposed algorithm

We now state the three-phase proposal this note is asked to develop, first as proposed, then — after the analysis of §§4–5 — in corrected form (§6).

3.1 Phase I: prolong, reduce, track

Prolong $A \cup \{P\}$ and auto-reduce until differential closure — every integrability condition (non-multiplicative prolongation, equivalently Δ -polynomial) reduces to zero. By the first correction of §2.4, this must be run as a completion loop (prolong $\rightarrow$ reduce $\rightarrow$ adjoin any nonzero residue, terminating by Ritt–Raudenbush), not as a single prolongation to a precomputed bound: for the combined system the constants-cutting conditions are the reduced derivatives $D^k \rho$ of the Ritt remainder, and no a-priori bound on k is available.

During every reduction step, record which reductor was used and how: *algebraic* steps divide by the reductor’s *initial*; *differential* steps (reduction against a proper prolongation θa) divide by its *separant*. In the algebraized picture the two cases unify: the initial of every proper prolongation θa is the separant of a , so the tracked set is simply “the initials of the (possibly prolonged) equations actually used.” The product of the tracked divisors is exactly Ritt’s denominator: the reduction identity is

$$h \cdot p \equiv r \pmod{(T^{[N]}), \quad h = \prod_i I_i^{d_i} \prod_j S_j^{e_j}, \quad (5)}$$

a polynomial identity, valid whether or not h vanishes; only the *conclusions* drawn from r (e.g. “ $r = 0$ hence p is a consequence”) require $h \neq 0$.

3.2 Phase II: recurse on the tracked inequations

For each tracked inequation h_i , split:

$$\text{Sol}(\Sigma) = \text{Sol}(\Sigma \cup \{h_i \neq 0\}) \sqcup \text{Sol}(\Sigma \cup \{h_i = 0\}),$$

keep $h_i \neq 0$ on the main branch, and recurse Phase I on the branch with h_i adjoined as an equation. Termination: adjoining h_i strictly enlarges the (radical differential) ideal, and strictly ascending chains of radical differential ideals terminate (Ritt–Raudenbush); when $h_i \in K[c]$ the split strictly descends the parameter dimension, the same descent that terminates Algorithm 2’s bad-locus recursion and CTD’s discriminant recursion. The output is a finite tree of systems ($E_i = 0$, $NE_i \neq 0$), each differentially closed (passive) relative to its inequations.

This is the Thomas/RG splitting discipline with lazy bookkeeping: Thomas also splits only on initials, separants, and squarefreeness conditions of equations it actually processes — not on arbitrary potential zero-divisors — so Phases I–II are best understood as a differential Thomas decomposition whose split set is generated *on demand* by the reduction trace. Two honest caveats. First, “used” must include the separants that license the *parametric-data freeness* of the final cell (each solved-form equation’s initial must be a unit for the Riquier existence theorem to apply); these are the separants of the final triangular set, which the completion loop does use, but an implementation that tracked only mid-reduction divisors and not final-form initials would emit cells on which Phase III’s existence semantics silently fail. Second, disjointness of the output requires carrying the complementary equations honestly ($h_i = 0$ branches), exactly as Thomas does; dropping them recovers Algorithm 1’s “main branch plus bad locus” shape.

3.3 Phase III as proposed — and why it cannot stand as stated

The proposal: in each equation of each leaf system, *group like terms by the independent variables and their algebraic extensions* (x, y, z, r with $r^2 = x^2 + y^2 + z^2$); because the equation must hold for all real values of the independents, set every grouped coefficient to zero; each coefficient involves constants and dependent variables; since a dependent variable need only take *some* value, *eliminate* the dependent variables from the coefficient system by Gröbner elimination and return the constant varieties admitting at least one solution in the dependents.

The organizing intuition — **independents universally quantified, dependents and constants existentially quantified** — is semantically correct, and §4 will keep it. But the operational recipe “collect over the independents first, then eliminate the dependents inside the coefficients” fails on both sides, and the failures are not edge cases:

Example 1 (over-constraint: the defining equation) *Apply the recipe to the ansatz equation $v - v_1x - v_2y - v_3z = 0$. Grouping by x, y, z gives four coefficient rows: v (the constant term), $-v_1$, $-v_2$, $-v_3$. Three rows contain no dependent variable at all, so the existential step has nothing to witness and the recipe forces $v_1 = v_2 = v_3 = 0$ — destroying a perfectly good defining equation of the ansatz. (The v -row is “witnessed” by $v = 0$, compounding the damage.)*

Example 2 (under-constraint: the frozen witness) *Apply the recipe to the combined toy system $A : \Psi'' - c\Psi$, $P : \Psi' - \Psi$ without the Phase-I closure (or, equivalently, apply it directly to Algorithm 1’s reduced remainder $r = \Psi' - \Psi$). No independent variable appears, so there is a single coefficient row $\Psi' - \Psi$, containing dependents only; the existential step asks whether some values (ψ_0, ψ_1) satisfy $\psi_1 - \psi_0 = 0$ — always yes. The recipe returns all of $\mathbb{C}$, but $\mathcal{E} = \{c = 1\}$ and $\mathcal{M} = \emptyset$. The recipe is not computing the existence locus: it has quantified over frozen values of Ψ, Ψ' , constant in x , whereas a solution’s jets vary with x — the constraint that kills $c \neq 1$ is the derivative of the imposed relation, $D(\Psi' - \Psi) = \Psi'' - \Psi' \rightarrow c\Psi - \Psi = (c - 1)\Psi$, which only differential closure supplies.*

Example 1 over-constrains ($\{v_1 = v_2 = v_3 = 0\}$ where the true answer is “no condition”); Example 2 under-constrains ($\mathbb{C}$ where the true answer is $\{1\}$). So the naive Phase III computes

a locus that is in general *incomparable* to both $\mathcal{M}$ and $\mathcal{E}$: it is the set of constants for which some x -constant assignment of the dependent symbols annihilates the equations identically in the independents — a hybrid with no differential meaning. The next two sections diagnose why, and what the correct discipline is.

4 The quantifier principle, made precise

4.1 The four classes of symbols

Fix a leaf system of Phase II: a differentially triangular set T with inequations NE , under a ranking $<$ on the jets. The non-constant symbols fall into four classes, and *each class admits exactly one sound projection operation*:

Definition 1 (symbol classes) *Relative to $(T, <)$:*

1. **Constants** c : the answer space. Never quantified away; the output lives in $K[c]$.
2. **Independent variables** x, y, z (and their algebraic extensions, here r with $r^2 = x^2 + y^2 + z^2$): universally quantified — an equation must hold at every point of some open set.
3. **Principal jets**: the leaders of T and all their derivatives. Each is determined by its equation (solved form; the initial is a unit by NE): its quantifier is irrelevant because it has a unique witness.
4. **Parametric jets**: the remaining jets — the free Taylor data of the family cut out by T (finite in number for a finite-type system). This class is the selector: quantify it universally and the projection computes membership; existentially, existence.

For the hydrogen ansatz under the paper's ranking $\Psi'' > \Psi' > \Psi > v > r > \text{constants} > x, y, z$: the principal jets are Ψ'', v, r (leaders), the spatial derivatives Ψ_x, Ψ_y, Ψ_z (leaders of the chain rules), and all their derivatives; the parametric jets are exactly Ψ, Ψ' . It is no accident that the paper's projection collects like terms in precisely x, y, z, r, Ψ, Ψ' — classes 2 and 4 — and in nothing else.

4.2 The two implementations of $\forall$, and the one of $\exists$

Principle 1 (operations, not just quantifiers) *Universal quantification is implemented by coefficient collection; existential quantification by elimination; determined symbols by substitution (triangular rewriting). Collection is sound only over symbols algebraically independent of one another and of everything retained; elimination over the jets is sound only after differential closure.*

We justify the three clauses in turn.

Collection ($\forall$). An equation $e \in K[c][x, y, z, r][\text{parametric jets}]$, containing no principal jet, vanishes for all real values of the independents and all admissible values of the parametric data iff every collected coefficient $p_\alpha(c)$ in (4) vanishes. Two ingredients:

- *Syntactic*: distinct power products are linearly independent over $K[c]$, so $e \equiv 0$ as a polynomial iff all $p_\alpha = 0$. For the algebraic extension r this requires e to be reduced modulo $r^2 - (x^2 + y^2 + z^2)$ first (r -degree ≤ 1); the reduced monomials are linearly independent as functions on the quadric. This is one of several places where “fully reduced” is load-bearing.

- *Semantic (density)*: vanishing *for all real* values is equivalent to identical vanishing because $\mathbb{R}^3$ is Zariski-dense in $\mathbb{C}^3$; vanishing for all admissible parametric data is equivalent because, on a cell with its inequations, the parametric Taylor data is freely assignable at every admissible base point (Riquier existence — for the hydrogen class, elementary second-order-ODE existence), so the realized values of $(x, y, z, r, \Psi, \Psi')$ fill a Zariski-dense subset of the quadric-times-jet space.

Note what collection is *not*: it is not elimination. One does not eliminate a universally quantified variable from an ideal — elimination computes the projection of a variety, which is the $\exists$ semantics. Collection is the linear-algebra operation dual to it, and this asymmetry is why the constants can stay in the coefficient field while the universal variables stay symbolic.

Elimination ($\exists$). For the existence reading, the object is the prolonged, algebraized, un-saturated ideal of the leaf system, $J = (T^{[N]}) \subseteq K[\text{jets}_{\leq N}, x, y, z, r, c]$, at closure order N . Lange-Hegermann’s Theorem 4.1 [10] (equivalently Lemma 3 of [3]) says that *for an involutive system*, every point of $V(J)$ respecting the inequations is the truncation of a genuine formal solution. Therefore

$$J_c = \left(J : (\Psi \cdot \prod_{h \in NE} h)^\infty \right) \cap K[c] \quad (6)$$

cuts out the Zariski closure of the cell’s contribution to $\mathcal{E}$: eliminate *all* the jets *and the base point* x, y, z, r (a solution needs only *some* point of analyticity — demanding all points would wrongly exclude, e.g., the Coulomb singularity $r = 0$), after quotienting out the zero section and localizing at the cell’s inequations. The universal “for all points near the base point” is carried entirely by the prolongation generators $\theta a \in T^{[N]}$: prolongation is the Taylor-basis implementation of $\forall$ -over-independents, dual to the monomial-basis implementation by collection. Example 2 is exactly what happens when this clause is ignored: without the generators $D^k(\Psi' - \Psi)$, the eliminated ideal knows nothing of $(c - 1)\Psi$ and the projection overshoots.

Substitution (determined). A principal jet is rewritten by its solved-form equation — this is what Ritt reduction *is*. Its equation is a definition, not a constraint row; its unique witness is supplied by the triangular structure. Feeding a defining equation into either projection operation is the category error of Example 1.

4.3 The two sound pipelines

Assembling Principle 1, there are exactly two sound orders of operations — one per quantifier assignment on the parametric jets — and the naive Phase III is sound in neither reading because it interleaves them [6, Finding 3: “exactly one $\forall$ algorithm and one $\exists$ algorithm”]:

	membership $\mathcal{M}$ (per cell)	existence $\mathcal{E}$ (per cell)
1. differential closure	full reduction of P by T	prolongation of $T \cup \{P\}$ to passivity
2. principal jets	substituted away by the reduction	eliminated (they are determined)
3. parametric jets	collected (universal)	eliminated (existential, after 1!)
4. independents x, y, z, r	collected (universal)	eliminated ($\exists$ base point; $\forall$ carried by 1)
5. constants	land in $K[c]$: the system $\{p_\alpha(c)\}$	land in $K[c]$: the ideal J_c of (6)

In the membership column no coefficient containing a dependent symbol is ever “set to zero”: collection continues *through* the parametric jets until the coefficients are pure constants. In the

existence column nothing is ever collected: every non-constant is eliminated, and the universal content lives in the prolongation generators. The trap is precisely the hybrid — collect over the independents, stop, and existentially witness the jet-carrying coefficients — because collection over x silently reclassifies every dependent symbol as a constant-in- x coefficient, which dependents are not (they are functions of x ; that is what “dependent” means).

Remark 1 (relation to comprehensive Gröbner systems and QE) *The $\forall\exists$ block structure over an algebraically closed field is a quantifier-elimination problem, and the tools align: the $\exists$ block is constructible-image computation (elimination ideals, per parameter cell — the comprehensive Gröbner system / Gröbner cover technology [18, 19]), while the $\forall$ block, because it is universal over a polynomial identity, degenerates to coefficient collection — linear, requiring no cell analysis and no real QE. This is why the whole pipeline stays inside Gröbner-land. The only real-vs-complex subtlety sits in the $\exists$ block: elimination certifies a complex witness; realness of solutions is a separate question the paper handles at the level of the final varieties.*

5 Resolution of the $v = v_1x + v_2y + v_3z$ puzzle

5.1 The distinguishing criterion

Criterion 1 *An equation may participate in independent-variable coefficient-zeroing if and only if it contains no principal jet — equivalently, iff it is fully reduced with respect to the cell’s triangular set — and the collection must then run over the independents and the parametric jets together, terminating in coefficients built from constants alone. An ansatz-defining equation never qualifies: its leader is a principal jet by definition — the equation is the solved form that defines that jet — so it enters the projection only as a rewrite rule (membership reading) or as an elimination generator (existence reading), never as a constraint row to be collected.*

Applied to the puzzle: in $v - v_1x - v_2y - v_3z = 0$ the symbol v is a dependent variable — itself a function of x, y, z — and under any admissible ranking it is the equation’s leader (it is the only jet present). The equation is therefore a definition of v , and Criterion 1 excludes it from collection in both readings:

- *Membership reading (Algorithm 1):* v is principal, so full reduction substitutes $v \mapsto v_1x + v_2y + v_3z (+ v_4r)$ everywhere in the remainder before the projection ever looks at it. The defining equation is consumed as a rewrite rule; v never appears among the collection variables — witness the paper’s collection set x, y, z, r, Ψ, Ψ' , from which v is conspicuously absent while the *parametric* jets Ψ, Ψ' are present.
- *Existence reading (bounded prolongation):* the equation and its prolongations $v_x - v_1, v_y - v_2, v_z - v_3, v_{xx}, \dots = 0$ enter the ideal J , and the v -jets are *eliminated*. The elimination ideal $(v - v_1x - v_2y - v_3z, v_x - v_1, \dots) \cap K[v_1, v_2, v_3]$ is zero: no condition on the constants, as it must be.

Either endpoint is safe. The naive Phase III fails because it stops the universal collection at the independents, leaving “coefficients” that still contain dependent symbols, and then treats those coefficients as objects that must vanish (rows without dependents — forcing $v_1 = v_2 = v_3 = 0$) or be witnessed by frozen values (rows with dependents — Example 2). Both failure modes are the same error seen from two sides: *collection over x, y, z presupposes that everything left in a coefficient is constant in x, y, z , and dependent variables are not.*

5.2 What distinguishes the defining equation from the reduced remainder

Both $v - v_1x - v_2y - v_3z$ and a reduced remainder r are elements of the same differential ring; both may contain jets; the remainder $r = \Psi' - \Psi$ of the toy example even has a jet *leader* (Ψ'). The distinction is not “contains a jet” but the *status of its jets relative to the triangular set doing the projecting*:

- the defining equation contains its own leader — a jet that is principal *for the ansatz* — so relative to A it is a definition;
- the fully reduced remainder contains, by the definition of full reduction, no leader of A and no derivative of one: every jet in it is *parametric* for A , i.e. free family data over which the membership question universally quantifies.

The ranking is what decides which jet of each equation is its leader, hence which jets are principal, hence which symbols the collection may treat as free. Change the ranking and the same polynomial can change roles — which is not a defect but the design: the ranking of the paper (§Ranking of [1]) is engineered exactly so that Ψ'' , the spatial derivatives, v , and r are the defined symbols and Ψ, Ψ' the free ones.

The “equal footing” aspiration is then realized without contradiction: adjoining P to A and decomposing means the remainder itself *becomes a defining equation* of the combined system — in the toy example, $\Psi' - \Psi$ acquires the leader Ψ' , turning Ψ' from parametric (for A) to principal (for $A \cup \{P\}$), and the completion then derives $(c - 1)\Psi$, whose *initial* $(c - 1)$ carries the constant condition as a Thomas split, not as a coefficient-zeroing. On an equal footing, constants conditions materialize as *initials of derived equations* and as *elimination ideals of cells* — never by zeroing jet-carrying coefficients. Coefficient-zeroing reappears only when, a cell having been fixed, one asks the membership question *of that cell* — at which point its defining equations act again as rewrite rules and Criterion 1 applies verbatim.

Remark 2 (why the physics tradition never met the puzzle) *The ansatz of [20] (and of the quasi-exact-solvability literature it implements) is explicit: $\psi = g(r)e^{y(r)}$ is substituted into the wave equation before any algebra happens, so every dependent symbol is eliminated by construction and raw power-equating is Criterion 1-compliant without anyone having to say so. The puzzle is a phenomenon of implicit (differentially triangular) ansätze, where definitions coexist with constraints in one system and must be told apart — by their leaders.*

5.3 The criterion tested on the partial-strata examples

Example A ($A : \Psi'' - c\Psi$, $P : \Psi' - \Psi$; one derivation). *Membership pipeline:* P is already fully reduced w.r.t. A (its leader Ψ' ranks below Ψ'' , and no derivative of Ψ'' appears); its jets Ψ, Ψ' are parametric for A ; collect over them: coefficients $+1, -1$; $\mathcal{M} = V(1, -1) = \emptyset$. ✓ (matches [2, Ex. 1]). *Existence pipeline:* close: $(A \cup \{P\})^{[2]} = \{\Psi'' - c\Psi, \Psi' - \Psi, \Psi'' - \Psi'\}$; auto-reduce: $\Psi'' - \Psi' \rightarrow c\Psi - \Psi' \rightarrow (c - 1)\Psi$, tracking no division (all initials are 1 until the derived equation appears); the derived equation’s initial $(c - 1)$ splits Phase-II-style: on $c \neq 1$ the cell forces $\Psi = 0$ (trivial; discarded by the $:\Psi^\infty$ quotient), on $c = 1$ the cell is $\{\Psi' = \Psi\}$ with parametric jet Ψ ; eliminating jets with $\Psi \neq 0$ leaves the ideal $(c - 1)$, i.e. $\mathcal{E} = \{1\}$. ✓ The dimension criterion agrees: the $c = 1$ cell has one parametric jet where A has two — a proper subfamily, hence a partial stratum, hence in $\mathcal{E} \setminus \mathcal{M}$.

Example B ($A : \Psi_{yy} - c\Psi$, $P : \Psi_x - \Psi$; **two derivations**). Here $A \cup \{P\}$ is already autoreduced *and coherent*: the Δ -polynomial of the leader pair (Ψ_x, Ψ_{yy}) reduces to zero, so Phase I terminates with no derived equation and Phase II never splits — one cell, equations $\{\Psi_x - \Psi, \Psi_{yy} - c\Psi\}$, parametric jets $\{\Psi, \Psi_y\}$. *Membership*: reduce P by A : remainder P itself; collect over the parametric data of A (which is infinite — A alone is not finite type; the collection over Ψ_x, Ψ suffices): coefficients $+1, -1$; $\mathcal{M} = \emptyset$. *Existence*: eliminate the jets from the closed cell with $\Psi \neq 0$: the prolongations $(\Psi_{xy} = \Psi_y, \Psi_{xyy} = \Psi_{yy} = c\Psi$ from both routes, ...) are consistent for every c ; the elimination ideal in $K[c]$ is (0) ; $\mathcal{E} = \mathbb{C}$ — every c is a partial stratum, the stratum being the subfamily $e^x(K_1 e^{\omega y} + K_2 e^{-\omega y})$. $\checkmark$ (matches [2, Ex. 2]). The dimension criterion again separates the readings: the cell’s parametric set $\{\Psi, \Psi_y\}$ is a collapse from A ’s infinite parametric set (two arbitrary functions $F(x), G(x)$ down to two constants), so no cell is dimension-preserving and $\mathcal{M} = \emptyset$ — even though the decomposition never split and the adjoined equation “reduced to nothing new,” confirming that membership cannot be read off the splitting combinatorics, only off the quantifier discipline (or dimensions).

The puzzle equation itself. Worked in §5: no condition on v_1, v_2, v_3 in either pipeline; $v_1 = v_2 = v_3 = 0$ arises only from the unsound hybrid. $\checkmark$

6 The corrected algorithm, and which locus it computes

6.1 Statement

Algorithm (prolongation–projection, corrected). *Input*: ansatz $A(c)$, PDE P , ranking $<$. *Output*: the existence locus $\mathcal{E}$ stratified into cells, with each cell’s membership status decided; hence both loci.

- I. Close and track.** Run the completion loop on $A \cup \{P\}$ (prolong $\rightarrow$ auto-reduce $\rightarrow$ adjoin nonzero residues) to passivity, tracking every initial/separant used as a divisor (5). No a-priori order bound is assumed (prolong-until-stable; the ansatz-only bound $N_0(A)$ is *not* valid for the combined system).
- II. Split and recurse.** For each tracked divisor h_i : keep $h_i \neq 0$ on the current branch; spawn a branch with $h_i = 0$ adjoined and recurse from I. Collect the leaves (T_j, NE_j) — finitely many, each passive.
- III_∃. Project each leaf onto the constants (existence).** Algebraize (jet $\mapsto$ indeterminate) at the leaf’s closure order; compute $J_c^{(j)} = (J_j : (\Psi \prod_{h \in NE_j} h)^\infty) \cap K[c]$, eliminating all jets and the base point. Then $\bar{\mathcal{E}} = \bigcup_j V(J_c^{(j)})$, each leaf contributing its stratum (bad-locus strata included — nothing was saturated by H_A globally).
- III_∀. Decide membership per leaf (optional selector).** On each leaf, either (a) fully reduce P by T_j and collect over independents *and* the leaf’s parametric jets, testing whether the leaf’s constant relations already imply every collected coefficient (radical membership, as implemented in `joca-rg-combined.sage`); or equivalently (b) test dimension preservation: the leaf’s parametric-derivative set equals the ansatz’s. Leaves passing are membership cells; their union recovers $\mathcal{M} = \mathcal{V}$; the rest are partial strata.

The changes from the proposal of §3 are exactly two, both forced by §4: Phase I is a completion loop rather than a one-shot prolongation to $N_0(A)$; and Phase III’s two quantifier moves are

unmixed — elimination of everything ($\text{III}_{\exists}$) or collection through everything non-constant ($\text{III}_{\forall}$), never collection-then-elimination on the same equation.

6.2 Which locus — and the selector

Phase $\text{III}_{\exists}$ computes, per leaf, the Zariski closure of that leaf’s contribution to the existence locus; the union over leaves is $\overline{\mathcal{E}} = \overline{\mathcal{V} \cup \{\text{partial strata}\}}$ [3, §2bis(b), superset remark]. It is a *new route to the joint-variety projection*, not to a third locus: by the two-pipelines analysis there is no third quantifier assignment to compute. What is genuinely selectable is the per-cell refinement:

- **Existence per cell** ($\text{III}_{\exists}$): keeps the bad-locus strata (hydrogen $E = -1/2$, $E = -1/8$ cells) that Algorithm 1’s $h \neq 0$ hypothesis fences off — the whole point of the equal-footing treatment.
- **Membership per cell** ($\text{III}_{\forall}$): the *dimension-preservation criterion* of [2, §The correct selection criterion] — a cell belongs to $\mathcal{V}$ iff the PDE removed nothing from the family, i.e. iff the cell’s parametric set is the ansatz’s. Lange-Hegermann’s differential counting polynomial [10] makes this effective by inspection of the cell (count the parametric derivatives); the radical-membership test of `joca-rg-combined.sage` is the implemented equivalent. Note the test must be the *radical-membership* form, not “re-reduce P by the cell’s chain and collect”: reduction against a chain’s bare equations does not saturate, and the hydrogen `joca-rg.sage` run shows the resulting projection re-admitting the spurious prime (a_0, a_1) while absorbing the genuine strata inside it [7].

So the corrected algorithm is a *selectable hybrid*: $\text{III}_{\exists}$ alone gives $\mathcal{E}$; $\text{III}_{\exists}$ filtered by $\text{III}_{\forall}$ gives $\mathcal{V}$; reporting both, with each partial stratum labeled by its cell, gives strictly more information than either endpoint algorithm — which is the honest way to state its value over Algorithm 1 (which cannot see the strata) and over a raw combined decomposition (which cannot tell the strata from $\mathcal{V}$ without exactly this selector).

6.3 Relation to the existing methods

Phases I–II are a differential Thomas decomposition with the split set generated lazily by the reduction trace — a strategy, not a new decomposition notion; the guarantees (disjointness, passivity of leaves) are Thomas’s [8], and any correctness claim should lean on that literature rather than re-prove it. Phase $\text{III}_{\exists}$ per leaf is the bounded-prolongation route of [3] run per cell (where its two known pathologies are mildest: on a cell the chain is triangular, the inequations are explicit, and the closure order is the cell’s own). Phase $\text{III}_{\forall}$ is Algorithm 1 run relative to a cell, with the saturation handled by radical membership. The prolong-to-order- i / algebraize / eliminate construction is classical (it is inside [14], there with the H -saturation we deliberately omit); the parametric splitting is comprehensive-Gröbner technology [18, 15]. What this note adds is not machinery but the *quantifier discipline* — the four-class table of Definition 1, the two-pipelines theorem-in-outline of §4, and Criterion 1 resolving the defining-equation puzzle — which is exactly the piece a correct implementation of Brent’s Phase III sketch was missing.

7 Open problems and next steps

1. **No a-priori closure order for $A \cup \{P\}$.** The ansatz bound $N_0(A) = 2$ does not bound the combined system; the constants-cutting conditions $D^k \rho$ enter at order $2 + k$ and the

terminating k is unknown for hydrogen (heuristically $\sim 4\text{--}6$; unmeasured). The pointwise probe `prolong-probe.sage` is written to lower-bound it empirically but has not yet produced data [6]. An a-priori bound in general would bear on the effectivity gap of the Ritt problem [13, 14] — do not expect one; `prolong-until-stable` is the algorithm.

2. **The parametric elimination cliff.** Direct $\text{III}_{\exists}$ over the eleven hydrogen constants as ring variables did not complete even over $\text{GF}(32003)$ at order 2. The recommended path is per-cell staging over the 29 already-computed Thomas cells, with the cheap membership filter ($\mathcal{M} \subseteq \mathcal{E}$) applied first; syzygy-based consistency extraction is the principled single-ideal fallback [6]. Until one of these lands, the corrected algorithm’s Phase $\text{III}_{\exists}$ is validated only on toy examples (§5.3).
3. **The “used inequations” completeness argument.** Phase II splits only on tracked divisors. The argument that this suffices — untracked initials/separants do not affect the reduction identities, and the leaf’s own solved-form initials are always used — should be written out against Bächler et al.’s simple-system conditions, in particular the squarefreeness splits (discriminants), which the tracking as described captures only when a gcd computation actually divides by them. Until then, treat Phases I–II as “Thomas with a lazy strategy, conjecturally split-minimal,” not as a proven refinement.
4. **Real witnesses.** $\text{III}_{\exists}$ certifies complex existence. For quantum-mechanical conclusions the real (and normalizability) questions are downstream filters on the resulting varieties, as in the paper’s treatment; a systematic real-existential layer (real QE / CAD) is unexplored territory for this problem class.
5. **Verification against the recorded runs.** The concrete check that would close the loop: run $\text{III}_{\exists}/\text{III}_{\forall}$ per cell on the 29 hydrogen cells and confirm (i) the membership cells’ union reproduces `joca.sage`’s five primes, (ii) the partial strata include the $E = -1/2$ and $E = -1/8$ cells with their recorded primes, and (iii) nothing else appears. The classifier of `joca-rg-combined.sage` already implements the $\text{III}_{\forall}$ test and validates on the toy examples; the hydrogen combined RG run itself does not terminate, which is precisely why the per-cell route matters.
6. **$\text{GF}(32003)$ vs $\mathbb{Q}$.** Any stratum destined for the paper needs a characteristic-zero confirmation of the modular computations.

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